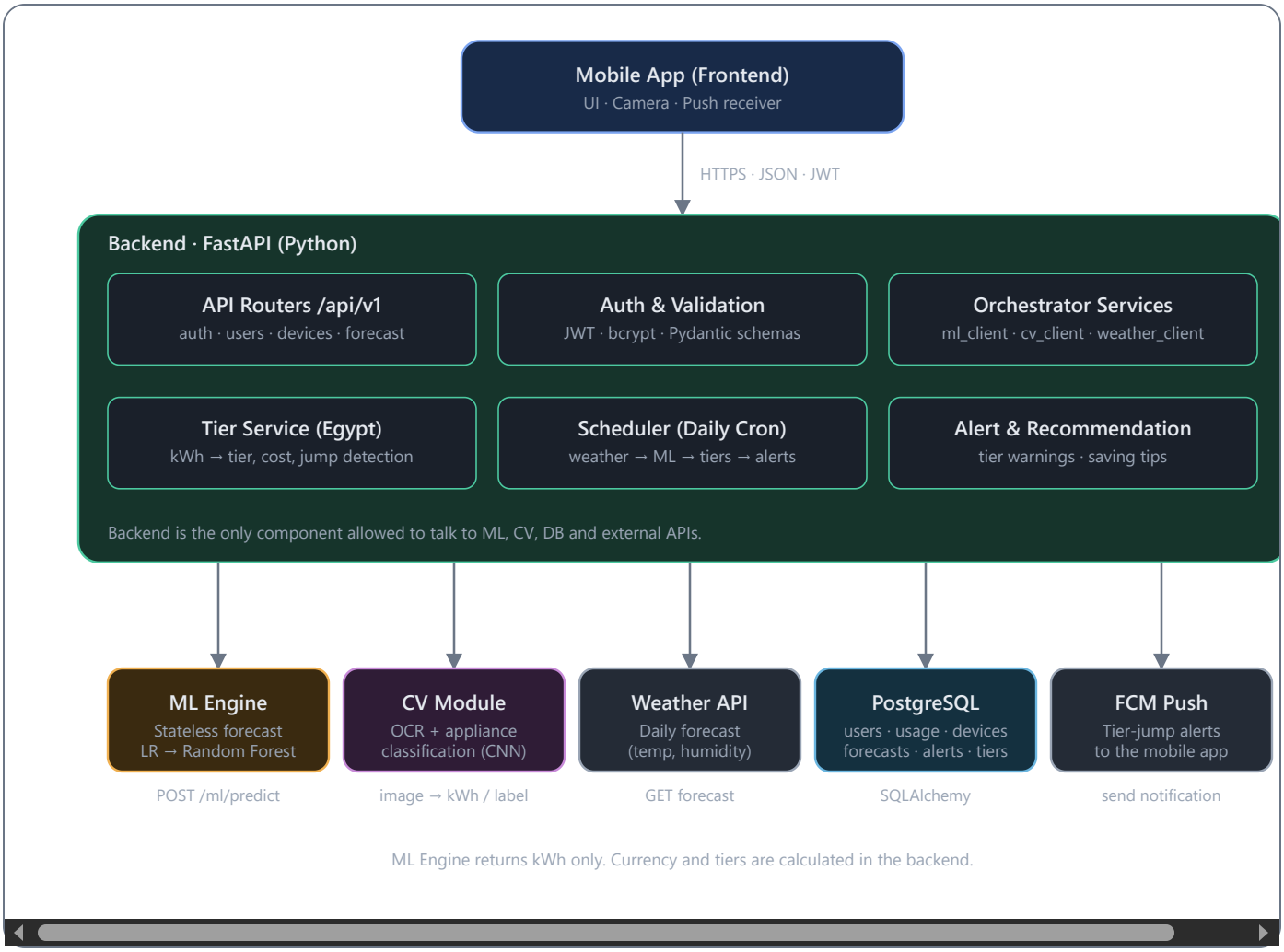
SECOS Backend Architecture

FastAPI backend as the single orchestrator between the mobile app, ML Engine, CV Module and external services.



Phase 1 · Cold start & calibration

1. App sends location, area, household size and an electricity-app screenshot.
2. Backend forwards the screenshot to the CV Module (OCR).
3. CV returns the historical kWh baseline.
4. Backend computes and stores the calibrated multiplier.

Phase 2 · Daily forecast & tier alerts

1. Cron job fetches the weather forecast.
2. Backend calls the ML Engine with profile, history and weather.
3. ML returns 7 to 14 days of predicted kWh.
4. Tier Service adds usage so far and projects the month.
5. If a tier jump is near, the backend stores an alert and sends a push.

Phase 3 · Appliance recognition

1. User photographs an appliance in the app.
2. Backend sends the image to the CV Module.

3. CV returns the device type and confidence.
4. Backend saves the device and returns saving tips.